

MUGANIZER

AI-POWERED MUSIC MANAGEMENT



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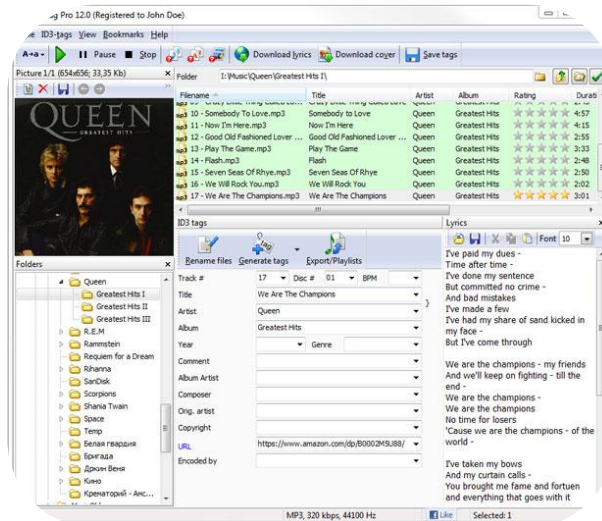
EGR404

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PROBLEM & MOTIVATION

- Local music libraries are difficult to organize
- Manual MP3 tagging is time consuming
- Playlists & searching is limited
- Current similar tools
 - Only in-app instead of local storage
 - Non-conversational
 - Cost money



Goal:

Create an AI-powered system that automates local music organization through natural conversation.

SYSTEM ARCHITECTURE

- User places untagged MP3 into Muganizer Inbox folder
- User interacts with GPT to tag a certain file (or multiple)
- GPT Actions send the request to the FastAPI server
- FastAPI routes the request to the Muganizer backend
- OpenAI API generates metadata
- Spotify API verifies metadata and retrieves cover (if applicable)
- Mutagen writes ID3 metadata tags
- Cover artwork is embedded
- Muganizer automatically organizes the file into:
 - Artist folder
 - Album folder
- Track metadata is stored in an SQLite database
- Semantic embeddings are created for AI-powered search & playlists
- GPT reports the results back to the user conversationally



GENERATIVE AI FUNCTIONALITY

OpenAI Metadata Generation

- From filename:
 - Title
 - Artist
 - Features
 - Genre
- From Librosa
 - Genre
 - Mood Tags
 - Descriptions

Semantic Search

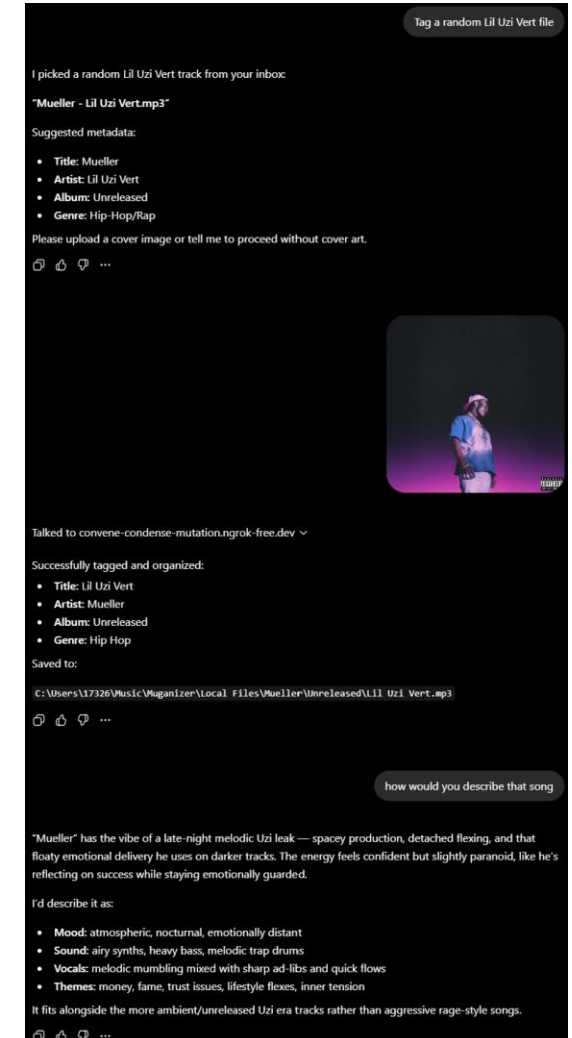
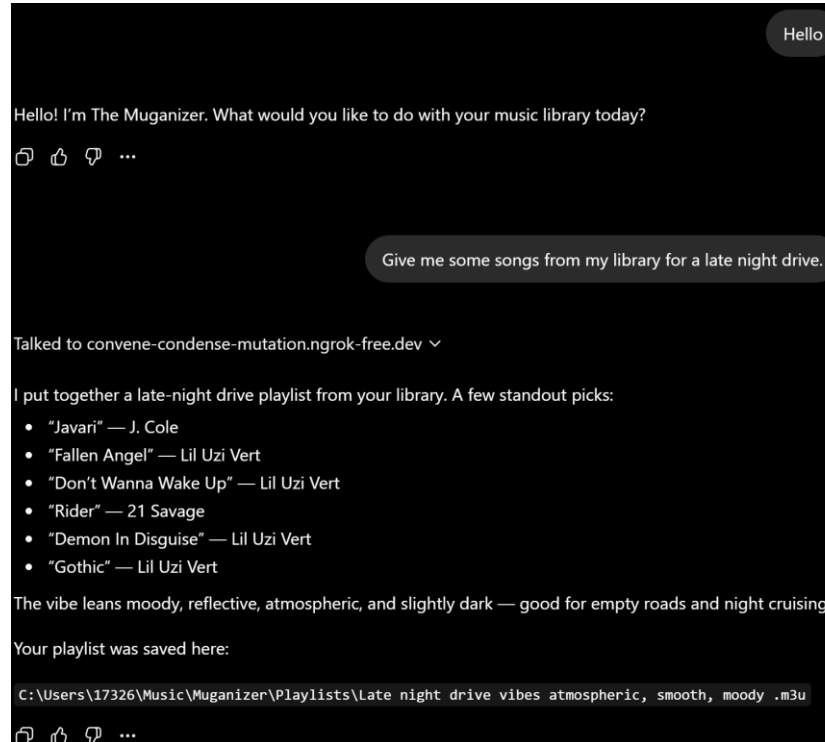
- Ex. “late night drive songs”

Playlist Generation

- Ex. “Make me a 40-song gym playlist”

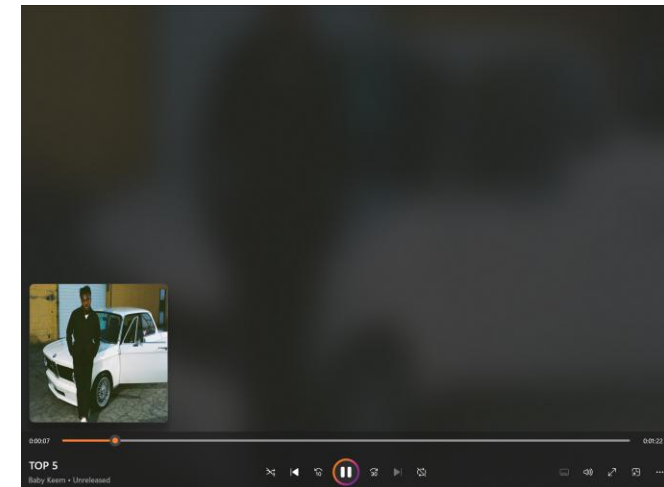
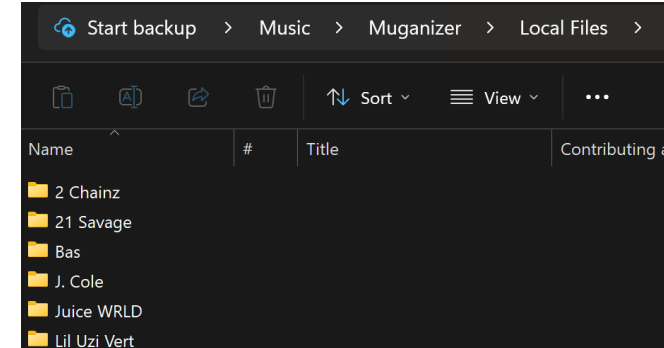
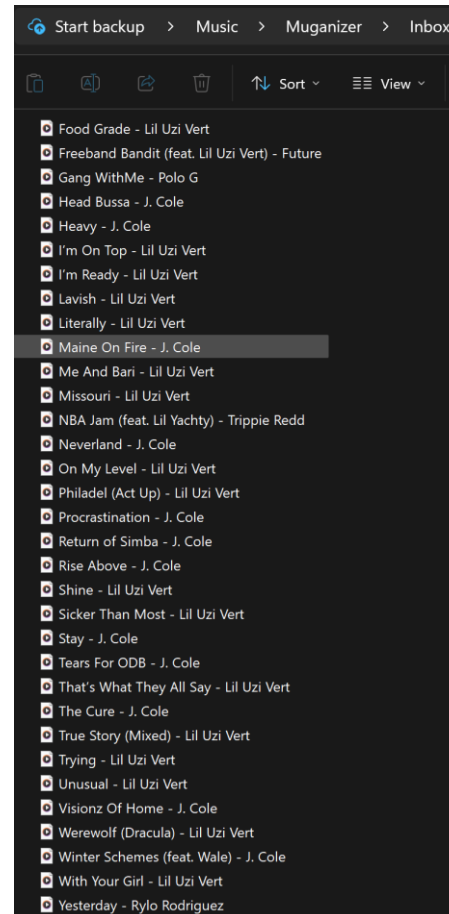
Conversational GPT Workflow

- Ex. “Tag a random file that has a J. Cole song”



END-TO-END WORKFLOW

1. User drops MP3 into Inbox
2. GPT detects file
3. AI generates metadata
4. User confirms metadata
5. Cover art downloaded
6. MP3 tagged automatically
7. File organized into library
8. Semantic embedding stored



RESULTS & FUTURE WORK

Results

- Fully operational minimum viable product
- Conversational interface
- Semantic search
- Playlist generation
- Tagging/organization

Future Work

- Fix cover on computer glitch (only works on phone atm)
- Integrate into web interface/application for customization
- More efficient tagging with large amounts of MP3's



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THANK YOU

